

Task 1 Dataset/partition

Task 1.1

Generate a dataset for future processing propose. File reequipments:

- CSV format
- Date column
- Approx at least 1GB
- Approx 100 columns

Example prompt:

Generate Python code for fake table for OLAP in form of one big table with 100 columns for hospital. Some of the columns should be text with symptoms of diseases, proposed treatment, test reports.

Task 1.2

Make an engineering decision about strategy for your data partitioning. Please explain your decision with evidence and business objective.

Data Partitioning vs Data Sharding

<https://medium.com/silenttech/data-partitioning-vs-data-sharding-4433288126f4>

<https://levelup.gitconnected.com/master-the-art-of-data-partitioning-for-system-design-interviews-a-complete-guide-b9d8075bc3cb>

Helpfull links:

<https://blog.stackademic.com/how-to-partition-data-across-shards-d1dfe0a57ee7>

Task 2 DuckDB

Instal DuckDB on your computer/cloud.

Upload your data as a table.

Develop SQL query to find out required charcteristics for data partitioning strategy.

Task 3 Hive Partitioning

Create the Hive Partitioning for your table (format **CSV**).

https://duckdb.org/docs/data/partitioning/hive_partitioning.html

Task 4

Create the Hive Partitioning for your table (format **PARQUET**).

Task 5: Delta Lake

Create Data lake for your data:

<https://duckdb.org/2024/06/10/delta.html>

Task 6: PySpark Words count

Step 1. Install PySpark on your enviroment.

<https://medium.com/@GalarnykMichael/install-spark-on-windows-pyspark-4498a5d8d66c>

<https://medium.com/@chenyumei8866/too-detailed-a-babysitting-level-tutorial-on-installing-pyspark-on-windows-97810462f850>

Step 2. In python select column with text data (text_col) from your BigData csv to be processed later.

3) In PySaprk save your selected column as Resilient Distributed Dataset (RDD),

- PySpark parallelize()

- PySpark `textFile()`
- PySpark `wholeTextFiles()`

<https://parmanand.medium.com/convert-an-rdd-from-a-text-file-into-a-dataframe-pyspark-66208b10f5d4>

4) Create PySpark code to count the words in your RDD.

- PySpark `flatMap()`

Some text about Python lambda:

https://www.w3schools.com/python/python_lambda.asp

<https://levelup.gitconnected.com/python-tutorial-31-python-lambda-functions-anonymous-functions-e8fad65e4f1f>

- PySpark `Map()`
- PySpark `reduceByKey()`

Some solution example:

<https://medium.com/@gulcanogundur/pyspark-word-count-b099106135a7>

<https://blog.devgenius.io/wordcount-using-pyspark-and-hdfs-2567367beb63>

Please do not use PySpark Dataframe.

Task 7: PySpark Dataframe

1. Read your CSV as a Dataframe DGF1
2. Write Dataframe DF1 as a Delta D1
3. Read your Delta D1 as a new Dataframe DF2
4. Run some SQL query for DF2

Task 8: DataBricks community edition tryout

Try creating your account, uploading the data and doing some processing (SQL + PySpark).

Task 9: